

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.SC. DIGITAL FORENSICS AND INFORMATION SECURITY

Subject Code: CTMSDFIS24 SI P2

Date: 16/12/2025

Subject Name: Cyber Security Audit and Compliance

Time: 2.00 PM TO 5.00 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Discuss one detailed case study highlighting the importance of audit in an organization.	08
(b)	Explain the different types of audits with proper scenarios.	08
(c)	Explain Audit, its importance, and the methods of assessment.	08
(d)	Describe IT audit approaches for Vendor and Third-Party Management.	08
Q.2	Attempt any three.	
(a)	Define the following terms with examples/scenarios: Policies, Framework, Guidelines, Rules, Laws.	08
(b)	Explain IT audit approaches for Physical Security.	08
(c)	List and elaborate the classification of critical control requirements for IT infrastructure audit and describe auditing standards of ISO/IEC 27001/2.	08
(d)	Explain IT audit approaches for Regulatory Compliance and Policy Compliance.	08
Q.3	Attempt any three.	
(a)	Explain IT audit approaches for Change Management, and Incident Response.	08
(b)	Explain IT audit approaches for Network & Infrastructure Security, Audit Trails & Logging.	08
(c)	What is Risk Analysis? Discuss Risk Assessment and Mitigation in detail.	08
(d)	Why is Business Continuity Planning required? Explain the BCP Life Cycle.	08
Q.4	Attempt any two.	
(a)	Explain ISO controls with brief discussion and list the IT domains for Cyber Audit with explanation of each domain.	07
(b)	Explain IT audit approaches for Access Controls.	07

- (c) What systems does an audit cover? Explain the steps involved in a security audit and threat analysis in IT security, including the scope, frequency, and objectives of an IT audit. **07**

Q.5

Attempt any two.

- (a) Explain how vulnerability, threat, and attack affect an IT security audit. Provide basic explanation of HIPAA, GDPR, and PCI-DSS. **07**
- (b) Answer the following (Workstation, LAN, WAN & Remote Access Domains): "one line answer for each point" **07**

Items found in User Domain

What is Workstation Domain?

Compliance requirements in Workstation Domain

Devices & components in Workstation Domain

Explain CIA Triad and how to maximize it in Workstation Domain

Devices & components in LAN/WAN Domain

Compliance and business drivers in LAN/WAN Domain

Penetration testing & configuration validation for ensuring LAN/WAN compliance

Compliance within Remote Access & Application Domain

Application server vulnerability & patch management for compliance

- (c) **Case Studies:** **07**
- (a) ShopEase disaster recovery scenario – Identify DR strategies to reduce downtime, data loss & maintain business reputation.
- (b) SlipCart hacking incident – Identify IT Act 2000/2008 sections applicable to:
- Website defacement
 - Spreading virus
 - DoS attack
 - Mention penalties for each.

END OF PAPER